

KisanGat - Schema Enhancements

(Recommended for Real-World Dairy Operations)

These tables should be added to the original database schema before development starts. They solve common real-world dairy business problems and prevent future database redesigns.

1. Milk Rate Management

Table: milk_rates

Purpose: Maintain historical milk pricing.

Problem Solved: If today's Cow Milk Rate is ₹42/L and next month it becomes ₹45/L, old records must still use ₹42/L.

Column	Data Type	Constraints
id	BIGINT	PK
milk_type	ENUM('COW','BUFFALO')	NOT NULL
purchase_rate	DECIMAL(10,2)	NOT NULL
selling_rate	DECIMAL(10,2)	NOT NULL
effective_from	DATE	NOT NULL
effective_to	DATE	NULL
is_active	BOOLEAN	DEFAULT TRUE
created_at	DATETIME	NOT NULL

Indexes

```
INDEX(milk_type)
INDEX(effective_from)
```

Business Rules

Only one active rate per milk type.

Historical rates must never be deleted.

2. Farmer Milk Rate Snapshot

Modification: milk_collections

Add Columns

Column	Type
rate_id	BIGINT FK
applied_rate	DECIMAL(10,2)

Relationship

rate_id  milk_rates.id

Reason:

Even if the milk rate changes later, the collection record retains the exact rate used.

3. Customer Milk Rate Snapshot

Modification: milk_sales

Add Columns

Column	Type
rate_id	BIGINT FK
applied_rate	DECIMAL(10,2)

Relationship

rate_id  milk_rates.id

Reason:

Historical bills remain accurate forever.

4. Dairy Information Settings

Table: dairy_settings

Purpose: Store dairy profile information.

Column	Type
id	BIGINT
dairy_name	VARCHAR(255)
owner_name	VARCHAR(255)
address	TEXT
phone	VARCHAR(20)
email	VARCHAR(255)
gst_number	VARCHAR(50)
logo_path	VARCHAR(500)
created_at	DATETIME
updated_at	DATETIME

Expected Rows:

Exactly one row.

5. Customer Ledger

Table: customer_ledger

Purpose: Track all customer financial transactions.

Column	Type
id	BIGINT
customer_id	BIGINT FK
transaction_date	DATETIME
transaction_type	ENUM('SALE','PAYMENT','ADJUSTMENT')
reference_id	BIGINT

Column	Type
debit_amount	DECIMAL(12,2)
credit_amount	DECIMAL(12,2)
running_balance	DECIMAL(12,2)
remarks	TEXT

Relationship

customer_id → customers.id

Benefits

Full customer account statement.

6. Farmer Ledger

Table: farmer_ledger

Purpose: Track all farmer transactions.

Column	Type
id	BIGINT
farmer_id	BIGINT FK
transaction_date	DATETIME
transaction_type	ENUM('COLLECTION','PAYMENT','ADJUSTMENT')
reference_id	BIGINT
debit_amount	DECIMAL(12,2)
credit_amount	DECIMAL(12,2)
running_balance	DECIMAL(12,2)
remarks	TEXT

Relationship

farmer_id → farmers.id

Benefits

Complete payment transparency.

7. Inventory Transactions

Table: inventory_transactions

Purpose: Maintain inventory movement history.

Column	Type
id	BIGINT
transaction_date	DATETIME
transaction_type	ENUM('COLLECTION','SALE','ADJUSTMENT','OPENING_STOCK')
milk_type	ENUM('COW','BUFFALO')
quantity	DECIMAL(12,2)
reference_id	BIGINT
remarks	TEXT

Benefits

Every inventory change becomes traceable.

Audit friendly.

8. Product Inventory Transactions

Table: product_inventory_transactions

Purpose: Track stock movement for dairy products.

Column	Type
id	BIGINT
product_id	BIGINT FK
transaction_date	DATETIME
transaction_type	ENUM('PRODUCTION','SALE','ADJUSTMENT')
quantity	DECIMAL(10,2)

Column	Type
balance_quantity	DECIMAL(10,2)
remarks	TEXT

Relationship

product_id  products.id

9. Farmer Bill Items

Table: farmer_bill_items

Purpose: Store detailed bill line items.

Column	Type
id	BIGINT
bill_id	BIGINT FK
collection_id	BIGINT FK
quantity_liters	DECIMAL(10,2)
fat_percentage	DECIMAL(5,2)
rate	DECIMAL(10,2)
amount	DECIMAL(12,2)

Relationships

bill_id  generated_bills.id

collection_id  milk_collections.id

10. Customer Bill Items

Table: customer_bill_items

Purpose: Store detailed customer billing entries.

Column	Type
id	BIGINT
bill_id	BIGINT FK
sale_id	BIGINT FK
quantity_liters	DECIMAL(10,2)
rate	DECIMAL(10,2)
amount	DECIMAL(12,2)

Relationships

bill_id → generated_bills.id

sale_id → milk_sales.id

11. Daily Business Summary

Table: daily_business_summary

Purpose: Fast dashboard loading.

Column	Type
id	BIGINT
summary_date	DATE
total_collection	DECIMAL(12,2)
total_sales	DECIMAL(12,2)
total_revenue	DECIMAL(12,2)
total_expenses	DECIMAL(12,2)
total_profit	DECIMAL(12,2)
pending_dues	DECIMAL(12,2)
inventory_balance	DECIMAL(12,2)

Unique Index

UNIQUE(summary_date)

Benefits

Dashboard loads instantly.

12. System Settings

Table: system_settings

Purpose: Store configurable application settings.

Column	Type
id	BIGINT
setting_key	VARCHAR(255)
setting_value	TEXT
description	TEXT

Examples

```
DEFAULT_PAYMENT_PERIOD
DEFAULT_REPORT_FORMAT
COMPANY_CURRENCY
MILK_UNIT
DATE_FORMAT
```

Final Database Count

Core Tables:

- users
- refresh_tokens
- farmers
- milk_collections
- farmer_payments
- customers
- milk_sales
- customer_payments
- milk_inventory
- products
- product_sales
- employees
- employee_attendance

- employee_salary_records
- expenses
- generated_bills
- report_exports
- audit_logs

Advanced Tables:

- milk_rates
- dairy_settings
- farmer_ledger
- customer_ledger
- inventory_transactions
- product_inventory_transactions
- farmer_bill_items
- customer_bill_items
- daily_business_summary
- system_settings

Total Tables: **28**

This schema is now robust enough for a real dairy business and can comfortably scale to multiple years of operations without major redesigns.